Student's assessment number:……

THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM ONE
032 CHEMISTRY

Time: 2:30 Hours **Thursday, 27th August 2026 a.m.**

Instructions

1. This paper consists of three sections, A, B, and C, with a total of ten questions.
2. Answer all questions in the spaces provided.
3. Section A carries 15 marks, Section B carries 70 marks, and Section C carries 15 marks.
4. All writing must be in blue or black ink except for diagrams, which must be in pencil.
5. Mobile phones and unauthorized materials are not allowed in the examination room.
6. Write your assessment number at the top right corner of every page.
7. Relative Atomic Masses (RAM): H=1, C=12, O=16, Na=23, Mg=24, S=32, Cl=35.5,

QUESTION NUMBER	MARKS	EXAMINER'S INITIAL
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
TOTAL		
CHECKERS INITIALS		

SECTION A (15 MARKS)

Answer **all** questions in this section.

1. For each of the items (i)–(x), choose the correct answer from the alternatives given and write its letter in the answer box provided.

(i) During a science exhibition, a student explains why chemistry is important in making medicines, fertilizers and cleaning products. Which statement best describes chemistry?

A. The study of living organisms only
B. The study of matter, its properties and the changes it undergoes
C. The study of the movement of planets
D. The study of rocks and minerals only

(ii) A student wants to determine the volume of water in a container more accurately than by using a beaker. Which apparatus is most appropriate?

A. Measuring cylinder
B. Test tube
C. Evaporating dish
D. Tripod stand

(iii) A student accidentally gets a small amount of a harmful chemical on the skin. What should be done first?

A. Rub the affected area with a dry cloth
B. Wash the affected area immediately with plenty of clean water
C. Cover the affected area with paper
D. Continue with the experiment

(iv) During a practical lesson, a teacher demonstrates that a liquid takes the shape of its container but maintains a definite volume. Which state of matter is being demonstrated?

A. Solid
B. Liquid
C. Gas
D. Flame

(v) Seypalm observes that ice changes into water when left in a warm place. This is an example of:

A. Freezing
B. Melting
C. Condensation
D. Sublimation

(vi) Which of the following is a **chemical change**?

A. Cutting paper into pieces
B. Melting candle wax

C. Rusting of iron
D. Dissolving sugar in water

(vii) A laboratory fire starts after a flammable substance catches fire. Which action is most appropriate?
A. Run away without informing anyone
B. Inform the teacher and use the appropriate fire-fighting equipment
C. Pour any available chemical onto the fire
D. Open all chemical containers nearby

(viii) Which apparatus is mainly used to transfer a small amount of a solid chemical?
A. Spatula
B. Thermometer
C. Funnel
D. Measuring cylinder

(ix) A student observes that perfume sprayed at one end of a classroom can eventually be detected at the other end. This illustrates:
A. Freezing
B. Diffusion of particles
C. Melting
D. Sedimentation

(x) Which of the following is the safest practice when heating a substance in a test tube?
A. Point the mouth of the test tube towards another student
B. Hold the test tube with bare fingers close to its bottom
C. Keep the mouth of the test tube pointing away from people
D. Seal the test tube tightly while heating

Answers

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)

2. Match each description in **List A** with the most appropriate term in **List B** by writing the correct letter in the table.

LIST A	LIST B
(i) A device used to measure temperature	A. Evaporation
(ii) A substance that burns and produces heat and light	B. Thermometer
(iii) Change of a liquid into a gas at its surface	C. First aid
(iv) Immediate assistance given to an injured person	D. Fuel
(v) A laboratory apparatus used to hold a small amount of a substance for heating	E. Test tube
	F. Condensation
	G. Tripod stand

Answers

List A	(i)	(ii)	(iii)	(iv)	(v)
List B					

SECTION B (70 MARKS)

Answer **all** questions in this section.

3. (a) Define the term chemistry.

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(b) Explain four ways in which chemistry contributes to the development of modern society.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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4. A student is preparing a school project on the treatment of water for domestic use. The project requires knowledge from several scientific fields.

(a) Name four disciplines that are related to chemistry and briefly explain one way in which each of the disciplines is related to chemistry.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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(b) State two benefits of learning chemistry as a Form One student.

(i) ……

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(ii) ……

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5. (a) During a practical lesson, students are required to measure liquids, heat substances, observe small objects and transfer chemicals. Identify the appropriate apparatus for each activity below.

(i) Measuring a known volume of liquid: .....…….

(ii) Heating a small amount of a solid or liquid: .....…….

(iii) Measuring temperature: .....……

(iv) Transferring a small amount of solid chemical: .....…….

(v) Observing very small objects that cannot be seen clearly by the naked eye: .....…….

(b) State five laboratory rules that students should observe when carrying out practical activities.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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(v) ……

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6. During a practical lesson, a student accidentally breaks a glass test tube and another student suffers a minor burn while handling heated apparatus.

   (a) State three safety measures for preventing accidents involving broken glassware.

   (i) ……

   ……

   (ii) ……

   ……

   (iii) ……

   ……

   (b) State two appropriate first-aid actions for a minor burn.

   (i) ……

   ……

   (ii) ……

   ……

   (c) Give two reasons why students should report laboratory accidents immediately to the teacher or laboratory technician.

   (i) ……

   ……

   (ii) ……

   ……

7. A fire starts in a school laboratory after a flammable substance comes into contact with a flame. The teacher immediately orders students to move to a safe place.

   (a) State three conditions necessary for fire to occur.

   (i) ……

   (ii) ……

   (iii) ……

   (b) Identify three suitable methods or materials that can be used to extinguish a fire, and state the principle by which they help to stop the fire.

   (i) ……

   ……

(ii) ……

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(iii) ……

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(c) State two precautions that should be observed when dealing with a laboratory fire.

(i) ……

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(ii) ……

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8. A student observes two different flames from a laboratory burner. One flame is yellow and produces soot, while the other is blue and produces much less soot.

(a) Name the two types of flames described.

(i) Yellow flame: .....……..
(ii) Blue flame: .....……..

(b) Give three differences between the two flames.

Yellow flame	Blue flame
(i)	
(ii)	
(iii)	

(c) State two situations in which the yellow flame is preferred in laboratory work.

(i) ……

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(ii) ……

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9. (a) A student places ice in a beaker and leaves it in a warm room. The ice changes into water and later some of the water changes into water vapour. Name the states of matter observed.

   (i) ……

   (ii) ……

   (iii) ……

   (b) Explain three differences between the three states of matter in terms of shape, volume and arrangement/movement of particles.

   (i) ……

   ……

   (ii) ……

   ……

   (iii) ……

   ……

   (d) Name the changes taking place when Ice changes into water: .....

SECTION C (15 MARKS)

Answer this question

10. At a school environmental club meeting, students investigate materials commonly found around their school. They observe an ice cube melting, water boiling, steam forming on a cool surface, a piece of paper being cut into small pieces, and an iron gate developing rust. Using these observations, answer the following:

   (a) Giving four points, distinguish between a physical change and a chemical change.

   (i) ……

   ……

   ……..

   (ii) ……

   ……

   ……

   (iii) ……

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(iv) ……

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(b) Classify each of the following observations made by the students as either a physical or chemical change. Give a reason for each classification.

(i) Ice cube melting:

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(ii) Water boiling:

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(iii) Iron gate rusting:

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(c) Give one everyday example of a chemical change other than any instance of a chemical change mentioned in (b)

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